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-- Databricks notebook source
-- MAGIC %md
-- MAGIC

-- COMMAND -----

-- MAGIC %md
-- MAGIC **Remove rows with missing customer_id from both datasets**

-- COMMAND -----

CREATE OR REPLACE TEMP VIEW customers_clean AS
SELECT *
FROM read_files('/Volumes/workspace/default/customers', format =>
'csv', header => true)
WHERE customer_id IS NOT NULL;

CREATE OR REPLACE TEMP VIEW orders_clean AS
SELECT *
FROM read_files('/Volumes/workspace/default/customers/orders.csv',
format => 'csv', header => true)
WHERE customer_id IS NOT NULL;

-- COMMAND -----

-- MAGIC %md
-- MAGIC ** Total order amount for each customer**

-- COMMAND -----

SELECT
    customer_id,
    SUM(total_amount) AS total_total_amount
FROM orders_clean
GROUP BY customer_id;

-- COMMAND -----

-- MAGIC %md
-- MAGIC **Top 3 customers by total spend**

-- COMMAND -----

SELECT
    customer_id,
    SUM(total_amount) AS total_spend
FROM orders_clean
GROUP BY customer_id
ORDER BY total_spend DESC
LIMIT 3;

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-- COMMAND -----

-- MAGIC %md

-- MAGIC **Customers with no orders**

-- COMMAND -----

-- MAGIC %md

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-- COMMAND -----

SELECT

 c.customer_id,

 c.*

FROM customers_clean c

LEFT ANTI JOIN orders_clean o

ON c.customer_id = o.customer_id;

-- COMMAND -----

-- MAGIC %md

-- MAGIC **City-wise total revenue**

-- COMMAND -----

SELECT

 c.city,

 SUM(o.total_amount) AS total_revenue

FROM customers_clean c

JOIN orders_clean o

ON c.customer_id = o.customer_id

GROUP BY c.city;

-- COMMAND -----

-- MAGIC %md

-- MAGIC **Average order amount per customer**

-- COMMAND -----

SELECT

 customer_id,

 AVG(total_amount) AS avg_total_amount

FROM orders_clean

GROUP BY customer_id;

-- COMMAND -----

-- MAGIC %md

-- MAGIC ****

-- COMMAND -----

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SELECT
    customer_id,
    COUNT(*) AS order_count
FROM orders_clean
GROUP BY customer_id
HAVING COUNT(*) > 1;
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-- COMMAND -----
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-- MAGIC %md
-- MAGIC Sort customers by total spend descending
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-- COMMAND -----
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```
-- Sort customers by total spend descending
SELECT
    c.customer_id,
    c.first_name,
    c.last_name,
    c.email,
    c.phone_number,
    c.address,
    c.city,
    c.state,
    c.zip_code,
    COALESCE(SUM(o.total_amount), 0) AS total_spend
FROM customers_clean c
LEFT JOIN orders_clean o
ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.first_name, c.last_name, c.email,
c.phone_number, c.address, c.city, c.state, c.zip_code
ORDER BY total_spend DESC;
```